

PRINTING + PROJECT AT A GLANCE

Use the CityScope as the physical thread of Phase 4. Keep the existing Science mission sequence and assessment roles. The model is an application object, not a separate giant grade.

PRINTING PLAN

Print one TEAM copy per group	Print one INDIVIDUAL copy per student
Pages 1, 2, 4, 5, 7, 8, 9, 11, 12, 13, 14	Pages 3, 6, 10
11 team pages	3 Math evidence pages

WHAT IS ACTUALLY GRADED?

Evidence	Primary use
Science	M1 formative · M2 formative · M3 stronger checkpoint · M4 formative · M5 major synthesis
Math	Page 3 Building Geometry · Page 6 Geometric Constructions · Page 10 Resource Mathematics
Shared project	Blueprint/model show application. Do not turn model quality into a separate large grade.
Not graded	Decoration · artistic skill · largest/tallest/strongest city · complexity for its own sake · energy/transport sophistication

SUGGESTED MODEL SIZE

Use one modest district base per team - approximately **poster-board size or smaller**, or a similar recycled cardboard base. The Primary Structural Test Building should be small enough to test safely on a desk. Keep the other two buildings simpler.

STORYLINE SAFEGUARD

Phase 4 answers **Could JCEC build this safely?** It does not answer **Should humans establish a long-term presence?** That remains the final Mission 2190 Council decision.

PAGE-BY-PAGE MARKING MAP

Page	Evidence type	Assessment / look-fors
1	Shared	Orientation only. Not graded.
2	Science · M1	Function → performance requirement → site evidence. Formative.
3	Math individual	Radius/diameter · circumference/ π · circle area/construction · 360° · triangle/parallelogram area.
4	Shared	Two concept options + reasoned choice. Completion/formative only.
5	Shared	Measured model plan; labelled dimensions and restrictions. Do not assess formal scale factor.
6	Math individual	Parallel · perpendicular · perpendicular bisector · angle bisector constructions.
7	Science · M2	External load → direction → internal force → likely failure. Formative.
8	Science · M3	Operational definitions · controlled variables · quantitative material/joint data.
9	Science · M3	Graph → property interpretation → application → limitation. Stronger checkpoint.
10	Math individual	Decimal operations + percent; optional decimal division.
11	Science · M4	Baseline stability evidence; foundation/support and mass distribution.
12	Science/shared	Two alternatives → selected redesign → comparable retest → evidence-based conclusion.
13	Science · M4	Safety Protocol + margin of safety. Formative.
14	Science · M5	Requirements/forces/materials/safety synthesis → Structural Authorization. Major synthesis.

MATERIALS + SETUP + LOW-MATERIAL FALLBACKS

MATERIALS TO GATHER

Category	Recommended shared supply
Planning	Graph paper / packet, pencils, rulers, calculators if normally used
Construction	Recycled corrugated cardboard, cardstock/paper, straws, masking tape, white glue, binder clips, string
Optional	Limited craft sticks, small reclaimed plastic/card pieces
Testing	Small safe masses/coins/washers, rulers, clips, string; 1–3 spring scales if already available
Avoid dependencies	Hot glue, saws, utility knives, structural kits, purchased model-city scenery

SETUP NOTES

- ☐ Store construction materials in one central JCEC Resource Depot rather than issuing full kits.
- ☐ Release major construction materials only after the M3 Material & Joint Recommendation.
- ☐ Standardize testing where the science requires comparison. Team CityScape models may be used for application, but keep a teacher/shared reference model or fallback dataset if models differ too much.
- ☐ Post teacher maximum loads and safe testing boundaries before M3/M4.
- ☐ Keep Chromebooks away from active build/testing zones where glue, masses or loose materials are present.

LOW-MATERIAL FALLBACKS

If unavailable	Use instead
Spring scales	One teacher demonstration or supplied force readings.
Several material types	Test 2–3 ordinary materials and pool class data.
Small masses	Coins/washers/counters in identical bags.
Enough building materials	One smaller Primary Building + paper/card versions of the other two.
Team model fails too early	Record why evidence is insufficient; switch to shared/teacher stability data and continue reasoning.
Construction time	Cut decoration/extra landmarks before cutting evidence, redesign or Team Records.

TESTING SAFETY + ASSESSMENT ROLES

TESTING SAFETY

- ☐ Use small desk-scale models and small safe loads.
- ☐ Set and display a teacher maximum load. Do not encourage testing to destruction.
- ☐ Do not reward the heaviest load held, tallest model or most dramatic failure.
- ☐ Keep fingers/faces out of the path of falling masses or collapsing structures.
- ☐ Use no hot glue, blades, saws or other unnecessary high-risk tools in the core pathway.
- ☐ Stop any unstable test immediately. Switch to shared/fallback evidence if needed.
- ☐ A worse redesign is valid evidence. Score the reasoning and comparable testing, not whether the fix 'wins'.

SCIENCE ASSESSMENT ROLES

Mission	Role	Primary packet evidence
M1	Formative	Page 2 · New Horizon Requirements Card
M2	Formative	Page 7 · Structural Force Map
M3	Stronger checkpoint	Pages 8–9 · fair test/data/graph/recommendation
M4	Formative	Pages 11–13 · stability/redesign/Safety Protocol
M5	Major synthesis	Page 14 · Structural Authorization

Recommended Phase 4 weighting remains **30% M3 / 55% M5 / 15% optional individual Structures & Forces check**, or approximately **35% M3 / 65% M5** without the check. Do not create an extra CityScape-model grade.

MATH EVIDENCE MAP

Page 3	Circle relationships/circumference/ π /area/construction/ 360° + triangle/parallelogram area
Page 6	Parallel, perpendicular, perpendicular-bisector and angle-bisector constructions
Page 10	Decimal multiplication/addition/subtraction + percent; optional decimal division

COMBINED MATH + SCIENCE PACING · WEEKS 1–3

Science keeps the approved **18 core classes + 7 flex periods**. CityScape uses selected Math periods alongside it. Do not use every Math class as build time.

WEEK 1 · REQUIREMENTS + GEOMETRY

Sci	Science	Math	Math / Project
S1	M1 · Function/forms/requirements	M1	Project Math launch + building dimensions
S2	M1 · Structure Detectives + culture/natural structures	M2	Building geometry calculations
S3	M1 · Site requirements	M3	Complete Math Evidence 1 + concept plan
S4	M2 · Mass vs force / loads		Regular Math / correction
S5	M2 · Internal forces + friction		
S6	M2 · Structural Force Map		
S7	FLEX · Force recovery		

WEEK 2 · CONSTRUCTIONS + MATERIAL SCIENCE

Sci	Science	Math	Math / Project
S8	M3 · Material properties / fair testing	M4	Geometric constructions instruction
S9	M3 · Material testing	M5	Parallel / perpendicular constructions
S10	M3 · Graph / joints / natural materials	M6	Bisectors + measured blueprint
S11	M3 · Material Recommendation	M7	Resource-budget mathematics
S12	FLEX · Checkpoint recovery	M8	Budget / decimal / percent evidence

WEEK 3 · BUILD / TEST / IMPROVE

Sci	Science	Math	Math / Project
S13	M4 · Stability / foundations / mass distribution		Regular Math
S14	M4 · Baseline test	M9	Optional data/graph connection or regular Math
S15	M4 · Alternatives / redesign / retest	M10	Model-dimension check / corrections
S16	M4 · Safety Protocol	M11	Budget update after redesign
S17	FLEX · Structures review		

COMBINED MATH + SCIENCE PACING · WEEKS 4–5

The final two weeks protect the Science synthesis and normal school interruptions. Do not stretch construction simply for decoration.

WEEK 4 · AUTHORIZATION

Sci	Science	Math	Math / Project
S18	M5 · Assemble structural system		Construction completion after required Math
S19	M5 · JCEC Design Review	M12	Final budget calculations
S20	M5 · Evidence + revision		Individual Math evidence recovery if needed
S21	M5 · Structural Authorization		
S22	FLEX · Authorization completion		

WEEK 5 · BUFFER

Sci	Science	Math	Math / Project
S23	FLEX · absence/recovery		
S24	FLEX · optional gallery review		
S25	FLEX · final buffer		

PACING SAFEGUARDS

- ☐ Cut decoration, extra landmarks, gallery work and model polish before cutting core evidence.
- ☐ Math Pages 3, 6 and 10 remain individual evidence even though the CityScape is collaborative.
- ☐ Only the Primary Structural Test Building completes the full Science testing/redesign cycle.
- ☐ Use Flex 22 for essential authorization completion; Flex 23–25 remain genuine recovery/buffer time.

QUICK MARKING LOOK-FORS + FINAL SETUP CHECK

SCIENCE LOOK-FORS

- M1** Requirements come from function/site evidence, not from a favourite design.
- M2** Force arrows/directions are sensible; students distinguish external load from internal force.
- M3** Fair test is controlled; graph uses quantitative evidence; recommendation matches a structural job; limitation is honest.
- M4** Two alternatives are considered; redesign responds to evidence; retest is comparable; margin of safety is conceptual.
- M5** Authorization uses evidence from multiple missions and balances function, reliability, safety, efficiency and environment.

MATH LOOK-FORS

- ☐ Page 3: correct relationships and area reasoning; perimeter is review/application only.
- ☐ Page 6: students show actual construction method/marks, not only visually plausible lines.
- ☐ Page 10: individual decimal and percent calculations are mathematically accurate and connected to team quantities.
- ☐ Do not assess formal scale factor. Page 5 is a measured model plan, not a scale-drawing outcome.

BEFORE THE PROJECT STARTS

- ☐ Print one 11-page Team Field File per group and Pages 3/6/10 per student.
- ☐ Confirm Phase 4 teams and assigned continents.
- ☐ Prepare central JCEC Resource Depot and price list.
- ☐ Choose approximate district-base size and safe testing limits.
- ☐ Prepare one shared/reference model or fallback dataset for M4 in case team models are not comparable.
- ☐ Have rulers, graph paper/packet pages, small safe loads and 1–3 spring scales if available.
- ☐ Tell students on Day 1: simple + evidence-based beats large + complicated.
- ☐ Keep decoration, extra landmarks and model polish outside grading.

Final operational rule: when time or materials become tight, cut decoration, extra examples, presentations and model polish before cutting the Science investigation, Math evidence pages, redesign/retest, or final Structural Authorization.